Data Architecture, Entity Analysis & ER Modeling — WariMitra

Document Identifier: WM-DAT-013

Classification: Enterprise Data Model & Entity-Relationship Specifications

Database Paradigm: Cloud Firestore NoSQL Document Database (`asia-south1`)

Target Audience: Data Architects, Database Administrators, Technical Business Analysts, Solution Architects

Source Evidence: `ARCHITECTURE.md:188-263`, `firestore.rules:30-185`, `functions/src/index.ts`

1. Data Architecture Strategy

WariMitra is architected around **Cloud Firestore**, a globally distributed, document-based NoSQL database optimized for high-concurrency real-time synchronization, mobile offline caching, and sub-500ms read latency.

1.1 Key Data Design Patterns:

1. **Document-Collection Hierarchies:** Deep relational joins are avoided; core business entities exist in root collections, while high-frequency child data (e.g., live GPS pings and clinic patient records) exist in isolated subcollections.
2. **Geospatial Indexing via Geohash:** Every geographic entity stores a 9-character string Geohash enabling hierarchical bounding-box range scans (`startAt` / `endAt`) across latitude/longitude surfaces without requiring heavy spatial database extensions.
3. **Optimized Aggregation Singletons:** Costly continuous full-collection scans are replaced by scheduled background aggregation functions that write to singleton documents (e.g., `analytics/dashboardMetrics`).
4. **Non-Destructive Audit Trail:** Operational collections preserve immutable history; `auditLogs` and `blockchainProofs` maintain append-only invariants enforced by database security rules.

2. Entity-Relationship Diagram (ERD)

MERMAID DIAGRAM

```
erDiagram
    USERS ||--o{ LIVE_LOCATION : broadcasts
    USERS ||--o{ PILGRIM_PROFILE : has
    USERS ||--o{ FAMILY_MEMBERSHIP : joins
    USERS ||--o{ SOS_ALERTS : initiates
    USERS ||--o{ MISSING_PERSONS : reports

    FAMILY_GROUPS ||--o{ FAMILY_MEMBERSHIP : contains

    SOS_ALERTS ||--o{ AUDIT_LOGS : triggers
    SOS_ALERTS ||--o{ BLOCKCHAIN_PROOFS : anchors

    MEDICAL_CAMPS ||--o{ PATIENT_ENCOUNTERS : admits

    POLICE_STATIONS ||--o{ PATROL_UNITS : operates

    TEMPLE_QUEUES ||--o{ DARSHAN_SLOTS : schedules

    USERS {
        string userId PK
        string email
        string firstName
        string lastName
        string userType
        string fcmToken
        timestamp createdAt
    }

    LIVE_LOCATION {
        string locationId PK
        float latitude
        float longitude
        int batteryLevel
        timestamp updatedAt
    }

    SOS_ALERTS {
        string alertId PK
        string userId FK
        string status
        float latitude
        float longitude
        string geohash
        string severity
        string idempotencyKey
        timestamp createdAt
    }

    MEDICAL_CAMPS {
        string campId PK
        string name
```

```
        float latitude
        float longitude
        string geohash
        int capacity
        int currentPatients
        int staffCount
        string status
    }

    PATIENT_ENCOUNTERS {
        string patientId PK
        string fullName
        int age
        string diagnosis
        string severity
        timestamp admittedAt
    }

    FAMILY_GROUPS {
        string groupId PK
        string name
        string ownerId FK
        string inviteCode
        array memberIds
        timestamp createdAt
    }

    MISSING_PERSONS {
        string reportId PK
        string reportedBy FK
        string name
        int age
        string photoUrl
        float lastSeenLatitude
        float lastSeenLongitude
        string status
        timestamp createdAt
    }

    BLOCKCHAIN_PROOFS {
        string proofId PK
        string entityId FK
        string eventType
        int eventVersion
        string canonicalHash
        string transactionHash
        string anchoringStatus
        timestamp createdAt
    }

    TEMPLE_QUEUES {
        string queueId PK
        string gateId
```

```
        string queueType
        int currentCount
        int waitTimeMinutes
        string status
    }
```

3. Core Business Entities & Usage Profiles

Entity 1: `users`

- **Business Meaning:** Master registry of all human actors interacting with the WariMitra platform.
- **Purpose:** Manages authentication identity, role-based authorization claims, and device push notification tokens.
- **Data Owner:** System Administration / User Account Holder.
- **Key Fields:** `userId`, `email`, `firstName`, `lastName`, `userType` (`pilgrim` | `volunteer` | `dindi_leader` | `medical_officer` | `police_officer` | `ngo` | `admin`), `fcmToken`, `createdAt`.
- **Relationships:** One-to-One with `pilgrims/profile`; One-to-Many with `liveLocation`, `sosAlerts`, and `missingPersons`.

Entity 2: `sosAlerts`

- **Business Meaning:** Official emergency distress incidents initiated by pilgrims or field personnel.
- **Purpose:** Represents the lifecycle of an emergency from trigger to field triage, dispatch, and final resolution.
- **Data Owner:** Police Command Center & Chief Medical Officer.
- **Key Fields:** `alertId`, `userId`, `status` (`pending` | `acknowledged` | `dispatched` | `resolved`), `latitude`, `longitude`, `geohash`, `severity` (`low` | `medium` | `high` | `critical`), `idempotencyKey`, `createdAt`, `createdAtClient`.
- **Indexes Enforced:** `(status, createdAt DESC)`, `(userId, createdAt DESC)`, `(severity, createdAt DESC)`.

Entity 3: `medicalCamps`

- **Business Meaning:** Physical medical relief tents, mobile clinics, and hospital triage posts along the pilgrimage highway.
- **Purpose:** Geospatial tracking of healthcare facilities, bed capacity, staffing levels, and patient

- **Purpose:** Geospatial tracking of healthcare facilities, bed capacity, staffing levels, and patient admissions.
- **Data Owner:** District Health Department / Medical Camp Superintendent.
- **Key Fields:** `campId`, `name`, `latitude`, `longitude`, `geohash`, `capacity`, `currentPatients`, `staffCount`, `status` (`active` | `inactive`).
- **Subcollection:** `patients/` recording individual clinical encounters.

Entity 4: `familyGroups`

- **Business Meaning:** Ad-hoc travel tracking cohorts formed by families or dindi contingents.
- **Purpose:** Enables private location-sharing and tracking among traveling members.
- **Data Owner:** Group Creator / Dindi Leader.
- **Key Fields:** `groupId`, `name`, `ownerId`, `inviteCode` (unique 6-character alphanumeric string), `memberIds` (array of UIDs), `createdAt`.

Entity 5: `missingPersons`

- **Business Meaning:** Formal reports of missing, lost, or separated pilgrims.
- **Purpose:** Captures biometric photographs and physical descriptions to aid police and volunteer searches.
- **Data Owner:** Solapur/Pune District Police Crime & Missing Persons Bureau.
- **Key Fields:** `reportId`, `name`, `age`, `category` (`child` | `elderly` | `adult`), `lastSeenLatitude`, `lastSeenLongitude`, `photoUrl`, `status` (`missing` | `investigating` | `found`), `reportedBy`, `createdAt`.

Entity 6: `blockchainProofs`

- **Business Meaning:** Cryptographic certification of critical state transitions in emergency incidents.
- **Purpose:** Provides mathematical proof that incident timestamps, locations, and response actions have not been altered.
- **Data Owner:** State Disaster Management Authority / Public Governance Audit.
- **Key Fields:** `proofId`, `eventType`, `entityId`, `eventVersion`, `canonicalHash` (SHA-256), `transactionHash` (Sepolia TxID or null), `anchoringStatus` (`CONFIRMED` | `LOCAL_PROOF_ONLY`), `snapshot` (immutable JSON), `createdAt`.

Entity 7: `analytics/dashboardMetrics` (Singleton Document)

- **Business Meaning:** Aggregated real-time macro telemetry for the Executive Command Center.

- **Purpose:** Eliminates database read costs by pre-computing system counts via a 1-minute background cron.
- **Key Fields:** `totalPilgrims`, `activeSOS`, `medicalCamps`, `avgQueueTime`, `updatedAt`.